ESE 3105 One-Shot Notes

Heat and Mass Transfer · KUET · past papers 2019, 2020, 2021, 2022, 2023 · answers from the course notes (Mahmudul sir, Mahfuz notes, Sohag sir) and the Cengel *Heat and Mass Transfer* textbook they follow

One-shot notes: ESE 3105 Heat and Mass Transfer

Modes and conduction equation (A1)

Conduction $Q = -kA\,dT/dx$ (Fourier); convection $Q = hA(T_s - T_\infty)$ (Newton); radiation $Q = \varepsilon\sigma A(T_s^4 - T_{sur}^4)$, $\sigma = 5.67\times10^{-8}$. $\alpha = k/\rho c_p$ (how fast heat spreads).

Cartesian: $\partial/\partial x(k\partial T/\partial x) + \ldots + \dot{e} = \rho c\,\partial T/\partial t$.

Cylinder: $(1/r)\partial/\partial r(kr\partial T/\partial r) + (1/r^2)\partial/\partial\varphi(k\partial T/\partial\varphi) + \partial/\partial z(k\partial T/\partial z) + \dot{e} = \rho c\partial T/\partial t$. Laplace/Poisson/diffusion forms.

BCs: temperature, flux, insulated, convection, radiation, interface. Generation = conversion, not creation. Contact resistance $R_c = \Delta T/(Q/A)$: roughness, pressure, hardness, fluid in gaps.

Resistances (A2)

Wall L/kA; film $1/hA$; cylinder $\ln(r_2/r_1)/2\pi kL$; sphere $(r_2 - r_1)/4\pi k r_1 r_2$; $Q = \Delta T/\Sigma R$; $1/U = 1/h_1 + L/k + 1/h_2$.

Wire with $\dot{e}$: $T_0 - T_s = \dot{e}r_0^2/4k$, $Q = \dot{e}\pi r_0^2 L$. $k = k_0(1 + \beta t) \Rightarrow k_m$ at mean temperature.

Answers: wall 85 °C/6300 W; wall + film 7.39 kW; brick 518 W; steam pipe 121 W/m, ΔT_{ins} 284 °C; wire 149.4/152.7 °C.

Critical radius (A3)

$R = \ln(r_2/r_1)/2\pi kL + 1/2\pi r_2 hL$; $dR/dr_2 = 0 \Rightarrow$ **$r_{cr} = k/h$** (cylinder), **$2k/h$** (sphere). Q is max at r_{cr}. Below r_{cr} insulation increases loss (good for electric wires); removing sub-critical insulation reduces loss.

Fins (A4)

$d^2\theta/dx^2 - m^2\theta = 0$, $m = \sqrt{(hp/kA_c)}$ (pin $\sqrt{(4h/kD)}$).

Insulated tip: $\theta/\theta_b = \cosh m(L - x)/\cosh mL$, $Q = \sqrt{(hpkA_c)}\theta_b \tanh mL$. Long: tanh = 1. Corrected $L_c = L + A_c/p$.

$\eta = \tanh mL/mL$; $\varepsilon = Q_{fin}/hA_b\theta_b$ (> 2 to be worth it).

Longer: more effective, less efficient. Rod 25.1 W; parabolic pin η 0.985, 1.82 W; motor: bare 356 W ≥ 350 W (no fins needed: flag).

Transient (A5)

$Bi = hL_c/k$, $L_c = V/A$ (sphere D/6, cylinder D/4, plate L/2); $Bi < 0.1 \Rightarrow$ lumped. $\theta/\theta_i = e^{-bt}$, $b = hA/\rho Vc_p$, $bt = Bi{\cdot}Fo$.

One-term: $\theta_0 = A_1 e^{-\lambda_1^2\tau}$ ($Bi = hr_0/k$); $Q/Q_{max} = 1 - 2\theta_0 J_1(\lambda_1)/\lambda_1$. Semi-infinite: $(T - T_s)/(T_i - T_s) = \text{erf}(x/2\sqrt{\alpha t})$; contact T_s weighted by $\sqrt{(k\rho c_p)}$.

Copper ball Bi 0.00075; brass 448 °C; shaft T_0 390 °C, 15.9 MJ/m; pipe depth 0.65 m.

Boiling and condensation (A6)

Curve: natural (< 5 °C) → nucleate (5-30, CHF ≈ 1 MW/m² at C) → transition (30-120) → film (> 120, Leidenfrost at D). Burnout: q > CHF jumps to film boiling at ~1000 °C; design q << CHF.

Drop-wise h 5-10× film-wise, but design for film.

Horizontal tube $h_1 = 0.729[g\rho_l(\rho_l - \rho_v)h^*_{fg}k^3/\mu\Delta TD]^{1/4}$, N tubes $/N^{1/4}$, $h^* = h_{fg} + 0.68c_p\Delta T$; vertical 0.943 with L. $h_{hor}/h_{vert} = 0.77(L/D)^{1/4}$.

33 tubes: 3510 W/m²K, 0.0664 kg/s per m; 10×10: 4875, 3.68 MW, 1.50 kg/s; 15×15: 7150, 0.278 kg/s/m; film boiling 940 W/m.

Convection basics (B1)

Convection = conduction at the wall + fluid motion; $h = -k(\partial T/\partial y)_0/\Delta T \approx k/\delta_t$, so h falls as δ_t grows.

Velocity layer (ν) vs thermal layer (α); $\delta/\delta_t \approx Pr^{1/3}$.

Turbulent: eddies, thin sublayer, steep gradient, high h.

$Re = VL/\nu$, $Pr = \nu/\alpha$, $Nu = hL/k$, $Gr = g\beta\Delta TL^3/\nu^2$, $Ra = GrPr$; $T_f = (T_s + T_\infty)/2$. Buckingham: 7 variables − 4 dimensions = 3 groups ⇒ $Nu = f(Re, Pr)$.

Boundary-layer analysis (B2)

Cubic profile ⇒ $\delta = 4.64x/\sqrt{Re_x}$ (Blasius 5.0); $C_{f,x} = 0.664/\sqrt{Re_x}$, $C_f = 1.328/\sqrt{Re_L}$. $\delta_t = 4.53x/Re^{1/2}Pr^{1/3}$ ⇒ $Nu_x = 0.332Re_x^{1/2}Pr^{1/3}$, average 0.664.

Energy eq: $u\partial T/\partial x + v\partial T/\partial y = \alpha\partial^2 T/\partial y^2$. Integral: $d/dx\int u(T_\infty - T)dy = \alpha(\partial T/\partial y)_0$. Colburn: $StPr^{2/3} = C_f/2$. e^{-ay} profile: $h = ka = 96.8$. $h_x = ax^{-0.1}$: $\bar{h}/h = 1.11$.

Flat-plate numericals (B3)

Laminar ($Re < 5\times10^5$) $Nu = 0.664Re^{0.5}Pr^{1/3}$; turbulent $0.037Re^{0.8}Pr^{1/3}$. Low P: only ρ changes. Unheated start: $\times[1 - (\xi/x)^{3/4}]^{-1/3}$.

Plastic sheet 437 + 548 = 985 W; 6 kPa air 569 W/m; heated after 12 cm: 262 W/m.

Internal flow (B4)

$Re < 2300$ laminar, $f = 64/Re$, $L_h = 0.05ReD$, $L_t = PrL_h$. Turbulent $Nu = 0.023Re^{0.8}Pr^n$ (0.4 heating, 0.3 cooling). $St = f/8$. Fully developed + uniform q: $\partial T/\partial x = dT_m/dx$ = const. Roughness affects only turbulent f.

$T_e = T_s - (T_s - T_i)e^{-hA/\dot{m}c_p}$; $Q = hA\Delta T_{lm}$; $\Delta P = f(L/D)\rho V^2/2$. Annulus h 8200; oil T_e 19.71 °C, 67.5 kW, 16.8 kW pump (inlet 20 °C assumed).

Tube banks (B5)

$V_{max} = S_TV/(S_T - D)$ (staggered: check $S_D < (S_T + D)/2$). In-line $Nu = 0.27Re^{0.63}Pr^{0.36}(Pr/Pr_s)^{0.25}$; staggered $0.35(S_T/S_L)^{0.2}Re^{0.6}$…; rows < 16 × F. $\Delta P = N_Lf\chi\rho V_{max}^2/2$.

Geothermal: h 92.2, T_e 29.1 °C, 24.9 kW/m, 21 Pa. In-line air: 69.4 °C, 87.7 kW, 0.41 kPa. Water staggered: 8190, 104 rows. Incropera: h 139, 19.8 kW/m.

Natural convection (B6)

$\beta = 1/T_f$; vertical $0.59Ra^{1/4}$ (L = height); horizontal $L_c = A/p$, hot up $0.54Ra^{1/4}$ ($0.15Ra^{1/3} > 10^7$), hot down $0.27Ra^{1/4}$; cylinder Churchill-Chu.

0.6 m plate: vertical 99 W (115 W Churchill-Chu), up 128 W (139 W), down 64 W. 8 cm pipe: 442 W.

Mass transfer (B7)

Fick: $J_A = -CD_{AB}dy_A/dx$; $D_{AB} \propto T^{3/2}/P$. Analogy: T↔C, k↔D, Pr↔Sc, Nu↔Sh. Homogeneous medium: no gradient, no diffusion. Absorption = bulk, adsorption = surface.

Equimolar: $N_A = CD(y_1 - y_2)/L$. Stagnant gas: $N_A = (CD/L)\ln[(1 - y_L)/(1 - y_0)]$. Sphere: $4\pi r_1r_2D\Delta C/(r_2 - r_1)$.

H_2 2.48×10^{-10} kg/s; toluene 5.49×10^{-7} kg/m²s; beaker 3.48×10^{-8} kg/s.

Abbreviations used in this pack

Every short form, written out. The answers also spell each one out the first time it appears.

Short form	Full meaning
BC	boundary condition
Bi	Biot number
CHF	critical heat flux
Fo	Fourier number
Gr	Grashof number
ID	inner diameter
kmol	kilomole
kPa	kilopascal
kW	kilowatt
LMTD	log mean temperature difference
Nu	Nusselt number
OD	outer diameter
Pr	Prandtl number
Ra	Rayleigh number
Re	Reynolds number
Sc	Schmidt number
Sh	Sherwood number
St	Stanton number